

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE**Paper 2 (Generated practice paper - Set 07)**

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours**Materials**

For this paper you may use:

- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
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9	
10	
11	
TOTAL	

Answer all questions in the spaces provided.

0 1 . 1

Which of the following is a feature of natural systems? Tick **one** box.

[1 mark]

- A Reactions that require high temperatures ☐
- B Reliance on non-renewable energy ☐
- C Waste products that are the raw materials for other processes ☐
- D Dispersal of elements so they cannot be re-used ☐

0 1 . 2

Explain why nitrogen fixation by bacteria is more sustainable than the Haber process.

[2 marks]

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0 1 . 3

Explain why incineration of wastes is less sustainable than natural decomposition.

[2 marks]

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5

0 2

Table 1 compares two exploited species.

Table 1

Feature	Anchovy	Blue whale
Age at sexual maturity	1 year	10 years
Offspring per female per year	Thousands of eggs	1 calf every 2-3 years
Lifespan	3-4 years	80+ years
Parental care	None	Extensive

0 2 . 1

Use Table 1 to identify which species is r-selected and which is k-selected, and explain the difference between the strategies.

[3 marks]

0 2 . 2

Explain why k-selected species are much more easily over-exploited than r-selected species.

[3 marks]

0 2 . 3

State the equation used to forecast a change in population size.

[1 mark]

0 2 . 4

Explain what is meant by **maximum sustainable yield** and how it relates to carrying capacity.

[3 marks]

10

03

Table 1 compares extensive oyster farming with intensive salmon farming.

Table 1

Feature	Extensive oyster farm	Intensive salmon farm
Food input	None - filter natural plankton	Fishmeal / plant pellets
Stocking density	Low, spaced on seabed	High, in sea cages / tanks
Breeding	Wild spat settle on cleared seabed or hatchery	Selected adults stripped of eggs and milt
Disease / pest control	Removal of predatory crabs	Antibiotics, pesticides, wrasse, removal of diseased fish
Yield per hectare	Low	Very high

03

1

Use Table 1 to explain the relative merits of extensive and intensive aquaculture.

[4 marks]

03

2

Describe the stages in breeding and rearing salmon in an intensive farm.

[3 marks]

03 . 3

Explain how shrimp farming has affected mangroves and how disease is controlled in shrimp lagoons.

[3 marks]

10

04 . 1

Explain why the ecosystem services provided by other species are often given a low conservation priority.

[2 marks]

04 . 2

Explain how soil organisms provide services that humans depend on.

[3 marks]

04 . 3

Explain why pollination by insects is more useful than wind pollination to plants growing in forests.

[2 marks]

0 4 . 4

Explain how seed dispersal by animals increases the survival of a plant's offspring.

[3 marks]

10

0 5

Table 1 shows typical energy values for an agroecosystem. Values are units of energy remaining from 100 000 units of solar energy.

Table 1

Stage	Energy / arbitrary units
Solar energy reaching the crop	100 000
Energy captured during photosynthesis	4 000
Energy stored in crop biomass	1 000

0 5 . 1

Calculate the percentage of the solar energy reaching the crop that is stored in crop biomass.

[1 mark]

Answer

%

0 5 . 2

Explain why so little of the solar energy is stored in the crop.

[3 marks]

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0 5 . 3

Explain why few food chains have more than four trophic levels.

[2 marks]

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0 5 . 4

Explain why rearing sheep or cattle is the best option for food production in upland and semi-arid areas.

[2 marks]

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0 5 . 5

Explain how pigs can increase the efficiency of a food production system.

[2 marks]

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10

06 . 1

Explain **four** properties of water that make it essential for life.

[4 marks]

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06 . 2

Describe how photosynthetic organisms changed the atmosphere and explain how this allowed life to colonise the land.

[4 marks]

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06 . 3

Explain how carbon sequestration by early photoautotrophs helped maintain a suitable temperature on Earth.

[2 marks]

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10

07

Table 1 shows climate data for two sites where a farmer is considering growing maize, a frost-sensitive crop that needs a thermal growing season of at least 120 days.

Table 1

Feature	Site A: valley floor	Site B: south-facing slope
Frost-free period / days	110	150
Days with mean temperature above 5 °C	160	175
Mean summer rainfall / mm	180	150
Gradient / %	1	8

07

1

Use Table 1 to explain which site is more suitable for maize.

[3 marks]

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07

2

Describe **three** ways in which temperature can be controlled to increase crop productivity.

[3 marks]

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07

3

Explain why keeping livestock warm increases the amount of food converted to growth.

[2 marks]

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07 . 4

Explain how day length affects livestock production and how it can be manipulated.

[2 marks]

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10

08 . 1

Explain why an electronic meter used to measure dissolved oxygen must be calibrated, and what should be done if the reading cannot be adjusted.

[2 marks]

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08 . 2

Give **three** types of indirect evidence of the presence of an animal and the information each can provide.

[3 marks]

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08 . 3

Explain how auditory monitoring can be used to study bats.

[2 marks]

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08 . 4

Explain **three** assumptions of the mark-release-recapture method that may not be met.

[3 marks]

10

09

The Russell formula describes changes in a fish stock: $S_2 = S_1 + (A + G) - (C + M)$, where S_1 and S_2 are the stock biomass at the start and end of the year, A = recruitment of young fish, G = growth, C = catch and M = natural mortality. Table 1 gives estimates for a cod stock.

Table 1

Term	Biomass / thousand tonnes
S_1	600
A	90
G	150
C	220
M	80

09 . 1

Calculate S_2 and state whether the stock is being fished sustainably.

[2 marks]**Answer**

thousand tonnes

09 . 2

Calculate the maximum catch that would have kept the stock constant.

[1 mark]

Answer thousand tonnes

09 . 3

Explain why the catch is the only term in the formula that is easy to measure, and why the others are difficult.

[3 marks]

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09 . 4

Describe how each of the following measures obtained from commercial catches can indicate overfishing.

[4 marks]

Catch per unit fishing effort

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Mean fish size and mean age

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10

Describe the conservation efforts used to protect tropical rainforests and evaluate their effectiveness.

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13

Extra space

10 . 2

State **one** reason why rainforest conservation is important to countries far from the tropics.

[1 mark]

10

Do not write outside the box

Write an essay on one of the following topics.

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Using tropical coral reefs as an example, discuss how the ecological features of a habitat determine both its importance and its vulnerability to human activities.

[25 marks]

OR

1 1 . 2

Discuss how feedback mechanisms, tipping points and diversity influence the stability of natural systems, and how human activities affect them.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

END OF QUESTIONS